ZRSense Board

ESP32-S3 Sense Robotics Carrier Board

Hardware Revision 1.0

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1. Overview

The **ZRSense Board** is a compact robotics carrier board designed around the **Seeed Studio XIAO ESP32-S3 Sense**. It combines sensing, lighting, motor control, servo outputs, switched outputs, power management, and expansion interfaces into a compact platform for educational robotics, hobby projects, and custom embedded control systems.

The board is intended to reduce the amount of hardware required to begin building a small robot. Rather than starting with a development board and designing separate circuits for distance sensing, motor control, lighting, power switching, and actuator connections, ZRSense provides these functions on a single board.

The XIAO ESP32-S3 Sense provides the processor, camera, wireless connectivity, and programmable control system, while ZRSense provides the hardware interfaces needed to build a compact robot or experimental platform.

Supporting firmware is available to simplify control of the integrated peripherals while still allowing users to develop custom applications and control systems.

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2. Key Features

- Designed for the Seeed Studio XIAO ESP32-S3 Sense
- Compact **35 mm × 22.5 mm** board
- Integrated VL53L5CX Time-of-Flight range sensor
- 12 individually addressable RGB LEDs arranged in a 3 × 4 array
- Single-channel bidirectional DC motor driver
- Two RC servo outputs
- Integrated white headlight
- MOSFET-controlled laser output
- Automatic switching between external 5 V power and a connected 1S LiPo/Li-ion battery
- External I²C expansion connector
- External UART connector
- Supporting firmware for integrated peripherals

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3. General Specifications

Parameter	Specification
Product	ZRSense Board
Hardware Revision	1.0
Main Controller	Seeed Studio XIAO ESP32-S3 Sense
Board Dimensions	35 mm × 22.5 mm
Primary Logic Voltage	3.3 V
External Power Input	5 V nominal
Battery Support	Single-cell LiPo / Li-ion
Power Source Selection	Automatic
Power Multiplexer	TPS2116DRLR
Distance Sensor	VL53L5CXV0GC/1
RGB LEDs	12
RGB LED Arrangement	3 × 4
DC Motor Channels	1
Servo Outputs	2
Laser Output	1
Headlight	Integrated white LED
I^2C Expansion	1 × 4-pin connector
UART Expansion	1 × 3-pin connector

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4. Functional Overview

The ZRSense Board uses the Seeed Studio XIAO ESP32-S3 Sense as its main controller. Integrated peripherals are connected directly to dedicated ESP32-S3 GPIO pins.

The board provides the following functions:

- Time-of-Flight ranging
- Addressable RGB lighting
- Bidirectional DC motor control
- Two servo outputs

- Headlight control
- Laser output control
- I^2C expansion
- UART communication
- Automatic power-source switching

[Insert Functional Block Diagram Here]

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5. ESP32-S3 GPIO Assignment

XIAO Pin	ESP32-S3 GPIO	ZRSense Function
D0	GPIO1	Laser control
D1	GPIO2	Addressable RGB LED data
D2	GPIO3	Pan servo
D3	GPIO4	Tilt servo
D4	GPIO5	I^2C SDA
D5	GPIO6	I^2C SCL
D6	GPIO43	UART TX
D7	GPIO44	UART RX
D8	GPIO7	Headlight control
D9	GPIO8	DC motor control
D10	GPIO9	DC motor control

The listed GPIOs are assigned to onboard functions or expansion interfaces.

Note: GPIO assignments should be verified against the final production schematic and board revision before relying on them for hardware compatibility.

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6. Integrated Time-of-Flight Range Sensor

The ZRSense Board includes an integrated **VL53L5CXV0GC/1 Time-of-Flight ranging sensor**.

The sensor communicates with the ESP32-S3 using the I^2C bus and can provide multizone distance measurements suitable for robotics applications.

Typical applications include:

- Obstacle detection
- Distance measurement
- Navigation
- Environmental awareness
- Object detection
- Robotic perception

Parameter	Specification
Sensor	VL53L5CXV0GC/1
Sensor Type	Time-of-Flight
Interface	I^2C
SDA GPIO	GPIO5
SCL GPIO	GPIO6
Maximum Zone Configuration	Up to 8 × 8
Maximum Zones	64

The sensor's effective range and performance depend on sensor configuration, target reflectivity, lighting conditions, and the operating environment.

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7. Addressable RGB LED Array

The ZRSense Board includes **12 individually addressable RGB LEDs** arranged in a **3 × 4 array**.

All LEDs are controlled through a single serial data connection from the ESP32-S3.

Parameter	Specification
LED Count	12
Arrangement	3 × 4
LED Type	WS2812B-2020-V6
Data GPIO	GPIO2
Control	Serial addressable LED protocol
Individual Pixel Control	Yes

The RGB LED array may be used for:

- Robot status indication
- Visual effects
- Direction indicators
- Sensor visualization
- Robot expressions
- Custom lighting effects

Power Note

The addressable LED array can consume significant current at high brightness.

When operating from a limited power source, firmware should limit LED brightness appropriately. The actual power consumption depends on the displayed color, brightness, and number of LEDs illuminated.

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8. DC Motor Driver

The ZRSense Board includes a **DRV8212DRLR single-channel H-bridge motor driver** for controlling one brushed DC motor.

The motor driver supports bidirectional motor operation and can be controlled using digital or PWM signals.

Parameter	Specification
Motor Driver	DRV8212DRLR
Motor Channels	1
Motor Type	Brushed DC
Control GPIO 1	GPIO8
Control GPIO 2	GPIO9
Motor Connector	CN6
Control	Bidirectional H-bridge

The motor driver is powered from the board's selected system power rail.

Motor Current Note

The DRV8212 integrated circuit has its own electrical ratings; however, these ratings should not automatically be considered the maximum continuous current rating of the complete ZRSense board.

Actual usable motor current depends on factors including:

- Power source capability
- Motor stall current
- Thermal conditions
- PCB design
- Connector capability
- System load

Users should select motors appropriate for the available system power.

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9. Servo Outputs

Two 3-pin RC servo outputs are provided for robotic actuators.

Typical applications include:

- Pan and tilt mechanisms
- Robot steering
- Grippers
- Sensor positioning
- Small robotic arms

CN2 — Tilt Servo

Pin	Signal
1	TILT_DRIVE
2	V_SHWD
3	GND

Control GPIO: GPIO4

CN3 — Pan Servo

Pin	Signal
1	PAN_DRIVE
2	V_SHWD
3	GND

Control GPIO: GPIO3

Parameter	Specification
Number of Outputs	2
Signal Type	RC servo PWM
Pan Servo GPIO	GPIO3
Tilt Servo GPIO	GPIO4
Power Rail	V_SHWD

Servo Power Note

The servo supply voltage follows the selected board power source.

When the board is externally powered, the servo voltage follows the board's external system supply. When operating from a connected battery, the servo voltage follows the battery-powered system rail.

Servo current requirements vary significantly. Users must ensure that the selected power source can safely supply the connected servos, particularly during startup or stall conditions.

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10. Headlight

The ZRSense Board includes an integrated white LED headlight.

The headlight can be switched or controlled using PWM for adjustable brightness.

Parameter	Specification
LED	Everlight EAHC2835WD4
LED Color	White
Control GPIO	GPIO7
Control	Digital or PWM
Function	Onboard headlight

The headlight may be used for:

- Forward illumination
- Robot lighting
- Camera illumination
- Visual status indication

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11. Laser Output

A dedicated MOSFET-controlled output is provided for connecting a compatible laser module.

The output uses an **AO3400A N-channel MOSFET** as a low-side switch.

Parameter	Specification
Control GPIO	GPIO1
Switching Device	AO3400A
Output Type	MOSFET low-side switched
Connector	CN1

CN1 — Laser Output

Pin	Signal
1	LZD_DRIVE
2	V_SHWD

The connected laser module receives positive power from V_SHWD while its negative connection is switched by the MOSFET.

Laser Safety

WARNING: LASERS MAY CAUSE SERIOUS EYE INJURY.

The user is responsible for selecting and operating an appropriate laser module and ensuring that the completed system complies with applicable safety requirements.

Never point a laser toward people, animals, vehicles, aircraft, or reflective surfaces.

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12. Power System

12.1 Automatic Power Selection

The ZRSense Board uses a **TPS2116DRLR power multiplexer** to automatically manage the available power sources.

The board supports:

- External 5 V power
- Single-cell LiPo or Li-ion battery power

The power system selects the available source and provides the selected power through the board's system power rail.

[Insert Power System Block Diagram Here]

12.2 External Power

External power is intended to be supplied as a regulated **5 V source**.

The TPS2116 power multiplexer supports an input voltage range of up to 5.5 V.

Recommended Specification

Parameter	Specification
Nominal External Supply	5 V
Power Management IC	TPS2116DRLR

Do not exceed the power input limits of the board or the connected XIAO ESP32-S3 Sense module.

12.3 Battery Power

A battery may be connected to CN5.

The board is designed for a compatible **single-cell rechargeable LiPo or Li-ion battery**.

Parameter	Specification
Battery Type	LiPo / Li-ion
Cell Count	1S
Nominal Voltage	3.7 V
Fully Charged Voltage	4.2 V
Battery Connector	CN5

CN5 — Battery Input

Pin	Signal
1	BAT
2	GND

Battery Warning

Do not connect a multi-cell battery pack directly to CN5.

Observe the correct connector polarity and battery voltage requirements.

Only use a battery appropriate for the XIAO ESP32-S3 Sense and ZRSense power system.

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13. I²C Expansion Interface

CN7 provides an external I²C interface for connecting compatible peripherals, including additional sensors and the ZRCore.

Parameter	Specification
Connector	CN7
Pin Count	4
Connector Pitch	1.25 mm
Interface	I²C
Logic Level	3.3 V
Power Output	5 V system supply

CN7 Pinout

Pin	Signal
1	GND
2	I²C SDA
3	I²C SCL
4	5V_IN

I²C Voltage Warning

Although the connector provides a 5 V power pin, the I²C communication signals are connected to the ESP32-S3's 3.3 V I²C bus.

Do not apply 5 V logic directly to SDA or SCL.

External devices must be electrically compatible with the board's 3.3 V I²C logic.

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14. UART Interface

CN4 provides a UART communication interface.

Typical applications include:

- GPS modules
- External microcontrollers
- Serial sensors
- Wireless communication modules
- Debugging
- Custom peripherals

Parameter	Specification
Connector	CN4
Pin Count	3
Connector Pitch	2.54 mm
Interface	UART
Logic Level	3.3 V

CN4 Pinout

Pin	Signal
1	GND
2	UART TX
3	UART RX

Signal	ESP32-S3 GPIO
UART TX	GPIO43

Signal	ESP32-S3 GPIO
UART RX	GPIO44

15. DC Motor Connector

CN6 provides the connection for one brushed DC motor.

Pin	Signal
1	M1_OUT_1
2	M1_OUT_2

Motor direction is controlled electronically by the H-bridge driver.

Connector: 2-pin, 1.25 mm pitch

16. Connector Summary

Connector	Function	Pin Count
CN1	Laser output	2
CN2	Tilt servo	3
CN3	Pan servo	3
CN4	UART	3
CN5	Battery input	2
CN6	DC motor output	2
CN7	I^2C expansion	4

[Insert Board Render with Connector Labels Here]

17. Mechanical Specifications

Parameter	Specification
Board Length	35 mm

Parameter	Specification
Board Width	22.5 mm
Hardware Revision	1.0

[Insert Dimensioned Mechanical Drawing Here]

The mechanical drawing should identify:

- Overall board dimensions
- Connector locations
- XIAO ESP32-S3 Sense position
- Sensor location
- Critical component locations
- Board thickness
- Mounting features, if applicable

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18. Typical Applications

The ZRSense Board is suitable for:

- Educational robotics
- Hobby robotics
- Small autonomous robots
- Pan/tilt camera systems
- Computer vision experiments
- Obstacle detection
- Distance sensing
- Embedded control projects
- Interactive projects
- Custom robotics platforms
- Experimental sensor systems

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19. Software Support

Supporting firmware is available to simplify access to the integrated ZRSense hardware.

Supported onboard functions include:

- VL53L5CX Time-of-Flight sensor
- Addressable RGB LED array
- DC motor driver

- Pan servo
- Tilt servo
- Headlight
- Laser output
- I^2C interface
- UART interface

The board may also be used with custom firmware developed using ESP32-S3-compatible development environments.

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20. Electrical and Power Considerations

The ZRSense Board combines multiple peripherals that may draw significant current depending on how the board is used.

Connected loads may include:

- DC motor
- Servo motors
- RGB LED array
- Headlight
- Laser module
- XIAO ESP32-S3 Sense
- Camera and wireless operation
- External I^2C devices

The total current consumption depends on the application and connected peripherals.

Users should select a power source capable of supporting the expected load and should consider peak current conditions, particularly when motors or servos start or stall.

Current Ratings

The datasheet does not specify a single maximum total system current or maximum continuous motor/servo current at this time.

The actual usable current depends on the complete power configuration, selected power source, connected loads, thermal conditions, and board-level operating conditions.

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21. Important Usage Notes

ESP32-S3 Logic

The ESP32-S3 uses 3.3 V logic.

Do not apply 5 V directly to ESP32-S3 GPIO pins, UART signals, or I²C communication lines.

Battery Safety

Use only a compatible single-cell rechargeable LiPo or Li-ion battery.

Do not:

- Connect a multi-cell battery directly to CN5
- Reverse battery polarity
- Short the battery terminals
- Use a damaged battery
- Leave charging batteries unattended
- Exceed the voltage limitations of the system

Motors and Servos

Motor and servo stall currents can be substantially higher than normal operating currents.

Verify that the selected power source is capable of supplying the expected peak load.

Addressable RGB LEDs

The RGB LED array can consume significant current at high brightness.

Consider limiting LED brightness in firmware when operating from battery power or a limited power source.

Laser Safety

A connected laser module may present a serious eye hazard.

The user is responsible for selecting and operating a laser module safely.

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22. Revision History

Revision	Date	Description
1.0	2026	Initial hardware release

23. Disclaimer

The ZRSense Board is intended for educational, hobby, and experimental use.

The user is responsible for verifying compatibility between the ZRSense Board and all connected motors, servos, batteries, lasers, sensors, and other peripherals.

Specifications may be updated as hardware characterization and validation continue.